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Domains of Recollection

Alan D. Baddeley
Medical Research Council Applied Psychology Unit
Cambridge, England

Four applied problem areas in long-term memory are explored using the concepts of encoding specificity and levels of processing. The areas are context-dependent memory in deep-sea divers, the problem of enhancing the ability to recognize faces, the attempt to understand amnesia, and finally the study of word finding in aphasic patients. All four areas produced some data that fitted neatly into existing concepts, but they also produced results that did not appear readily explicable. In order to account for these discrepant results, three new concepts are introduced. First, the concept of levels is replaced by a more general concept of domains of processing. Second, two types of contextual effect are distinguished: interactive context, which influences the memory trace directly by affecting the way in which the stimulus is encoded, and independent context, in which stimulus and context are both processed but do not interact. The first type of context is assumed to affect both recall and recognition, whereas the second is readily detected only in recall. Third, a concept of recollection is proposed; this supplements the concept of an automatically cued process of retrieval by assuming active processes of search and evaluation. It is suggested that the study of applied memory problems offers a valuable supplement to more traditional laboratory-bound approaches to cognitive psychology.

Although cognitive psychology developed primarily within the experimental laboratory, it is showing an increasing tendency to extend its influence into the outside world. There are two major reasons for this development.

The first of these stems from the search for ecological validity, driven by the conviction that cognitive psychology should be concerned with understanding human behavior in general. This is often coupled with the suspicion that some of the laboratory studies of the 1960s may be telling us a great deal about the way college students play the particular cognitive games presented by our conventional laboratory tasks, but little else. Fortunately, many of the theoretical concepts of cognitive psychology appear sufficiently robust to be applicable outside the laboratory, and although the results one obtains in the outside world are not always predictable from laboratory studies, nonetheless the relation does seem to be sufficiently close to be encouraging (Baddeley, 1981).

The second reason for an extension of cognitive psychology beyond laboratory-based theoretical studies stems from a growing concern with applied problems, influenced in part no doubt by the changed economic and political climate. Research funds are tighter, and the need to offer justification for research to the nonscientist has become considerably more pressing. Furthermore, with the contraction of the university system, graduate students are increasingly finding their future in applied research rather than in teaching.

In the past, applied and academic research have often proceeded in parallel with little or no interaction. If cognitive psychology is to continue to flourish, however, it is important that cross-fertilization occur between pure and applied research. The present paper argues that such links are both possible and healthy and supports this claim by describing three concepts developed from applied problems that are applicable to central issues in the theoretical study of human memory. The

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Requests for reprints should be sent to Alan Baddeley, Medical Research Council Applied Psychology Unit, 15 Chaucer Road, Cambridge CB2 2EF, England.

way these developed from applied problems will be outlined, and their relation to concurrent developments in more laboratory-based studies of memory will be discussed.

A major aim of this article is to argue that applied problems can give rise to potentially important theoretical insights. Hence, I suggest that applied research can provide a fruitful avenue to the discovery of new basic phenomena and novel concepts with which to explain them. The relation between pure and applied research in this view then is one of partnership between two methods, both converging on a similar set of theoretical concepts. In order to make this case, I propose first to present the novel concepts, then to demonstrate their usefulness in tackling a range of applied questions, and finally to relate them to current theoretical work in human memory.

The Concepts

The Starting Point

The two theoretical approaches I attempt to apply are Craik and Lockhart's levels-of-processing and Tulving's encoding-specificity hypothesis. In their levels-of-processing approach to memory, Craik and Lockhart (1972) suggest that a great deal of evidence can be explained on the assumption that the amount of information recalled is a function of the type of processing required during learning. More specifically, they assume that information processing is hierarchical, with an incoming stimulus such as a written word being processed first in terms of its visual and orthographic characteristics, then in terms of its phonological characteristics, and subsequently in terms of its meaning. Orthographic processing is assumed to be shallow and to give rise to poor subsequent retention; the slightly deeper phonological code is more durable, whereas the even deeper semantic code is the most durable.

The encoding-specificity hypothesis, which was subsequently elevated by its proposers to the level of a principle, states that "specific encoding operations performed on what is perceived determine what is stored, and what is stored determines what retrieval cues are effective in providing access to what is stored" (Tulving & Thomson, 1973, p. 369). The principle was devised to account for the role of retrieval cues in memory, a retrieval cue being a stimulus, in most experiments presented during learning, which proves to enhance retrieval when re-presented at recall. Although direct evidence for the encoding-specificity hypothesis has been derived primarily from the cuing paradigm, it has been generalized to cover "all known phenomena of episodic memory and retrieval" (Tulving & Thomson, 1973, p. 370).

Both levels and encoding specificity have subsequently been extended and modified into a richer and more complex framework. The three concepts that follow can perhaps best be conceptualized as part of this process of conceptual elaboration and enrichment.

Processing Domains

It is suggested that the concept of levels of processing be replaced by the more general concept of processing domains. A domain is an area of memory within which there are extensive associative links and connections. Such areas of memory will often be associated with particular cognitive processes that may well be separable functionally and possibly, indeed, anatomically. Hence, the orthographic processing of a word may be carried out in a separate cognitive system from its semantic processing. Bearing this in mind, it is tempting to identify domains with the codes employed by clearly separable processing systems. However, although this may be the case for relatively peripheral domains such as those involved in reading, this definition would be far too constricting when applied to long-term memory, where it seems reasonable to identify a wide range of domains and subdomains, without necessarily implying that these utilize uniquely different codes. Indeed, even such a clear distinction as that between the domains involved in visual and verbal processing cannot be attributed with any degree of confidence to differential codes. The available evidence clearly indicates differences between visual and verbal memory but is quite compatible with the conceptualization of a common abstract long-term code. Differences between visual and verbal memory could be reflected in such a system by the type of material stored in this abstract code, and also by the preferred means of manipulating such material during storage and output (Baddeley & Lieberman, 1980; Kosslyn, 1980).

A processing domain, then, is essentially an area of memory containing a rich network of links and connections; such domains will often be associated with separable cognitive processing systems, but this need not be the case. Finally, although the rich interconnections are referred to as associations, it is not intended to tie the concept to any particular associationist assumptions. The concepts would fit equally well with a holographic or resonance model, provided such a model is capable of reflecting the organization of material in memory. Domains are connected to other domains and may be divided into subdomains, so the distinction is not absolute although it is, however, a useful one.

The concept of domains differs from the concept of levels in a number of ways. Unlike levels, it does not assume a linear sequence or a hierarchy of processes, because information may be processed within a number of domains simultaneously, in a parallel heterarchical manner. In the absence of a hierarchy, it does not have necessary implications for the durability of memory traces within any given domain, although it is likely that processing within different domains will have different mnemonic consequences. Similarly, the concept of a domain does not have any necessary implications for whether maintenance rehearsal will lead to learning. In short, the concept of a domain is a very much more general theoretical structure than that assumed by levels of processing. I should point out, however, that most of the stronger assumptions made by levels of processing have proved to be either unjustified or else explicable in terms of other more general principles, such as elaboration or cue overload, both of which are applicable within the more general concept of a domain (Baddeley, 1978).

To what extent is it worth retaining such a pale shadow of the levels-of-processing concept? As I hope to demonstrate later, it is extremely useful to have a concept that allows one to express the fact that certain types of information are closely related and interact strongly, whereas other types are relatively separate. An obvious case in point is that of visual and verbal memory. It is clear that words can evoke visual images, and visual stimuli suggest words, but nonetheless, the abundant evidence for separate visual and verbal coding indicates that visual and verbal memory can usefully be conceptualized as distinct but related domains (Milner, 1971; Paivio, 1971; Woodhead & Baddeley, 1981).

Reference to a given domain does not of course imply a single monolithic and unitary system. For example, Morton (1979) has distinguished three subcomponents of the lexical domain, associating each with a separate logogen system, one for words presented visually, one for aurally presented words, and a third serving as an output logogen system. Clearly, all three are likely to be strongly interconnected, but their functioning can be influenced independently by different types of priming. The concept of a domain is intended to be relatively atheoretical but at the same time may be mapped onto more detailed and less theoretically neutral approaches to perception and memory such as those of Morton.

A concept such as domains of processing is not in any simple sense testable, nor is there any one experiment one can carry out that will tell us whether the concept is useful or useless. There is little doubt, however, that we do need to explore the interrelation among lexical, semantic, and episodic representations in memory, and the test of a concept such as domains of processing lies in whether it proves useful in guiding and encouraging the exploration of such relations without allowing strong prior assumptions to constrain investigation unduly.

Interactive and Independent Context

It is suggested that context may influence learning and subsequent memory in two separate ways. The most dramatic effect of context will occur when the context determines the way an item is encoded. In the classic verbal-memory studies by Tulving and his collaborators (e.g., Tulving & Thomson, 1973), the word to be remembered (TBR) is accompanied by a low-frequency associate; for example, the TBR word *cold* might be accompanied by the word *ground.* Under such conditions, subjects may be more likely to recall the TBR word given the retrieval cue *ground* than they would be to recognize the word *cold* when they have generated it in another context, for example, as an associate of the word *hot* (Tulving & Thomson, 1973). It is suggested that this is because the word *ground* actively modifies the way in which subjects interpret the TBR word *cold* so that what they store is an experience created by the interaction of the two, for instance, the thought of a grave or perhaps a field under a frost. Interactive context therefore operates by changing what is stored. One consequence of this is that interactively coded context will influence both recall and recognition.

Context need not, however, always be interactive. The subject may encode both the TBR word and the cue quite separately, particularly if they come from very different domains. Environmental context generally appears to operate in this way; subjects encode the word at the same time as they encode its environmental surroundings. Reinstating the environment will enhance recall, but in contrast to interactive context, which also influences recognition, independent context appears to leave recognition memory unaffected.

As we shall see later, the distinction between interactive and independent context applies not only to the role of environmental context but also to results in the role of semantic context in face recognition (Baddeley & Woodhead, 1982), and under certain conditions to verbal memory (Gardiner & Tulving, 1980).

The distinction between interactive and independent context was almost certainly influenced by Garner's work on perception (Garner, 1962, 1976) and by an unpublished paper by Hewitt (Note 1). Hewitt proposed the distinction between intrinsic and extrinsic context, with intrinsic context applying to characteristics of the stimulus that subjects need to encode in order to carry out the task they are set in contrast to extrinsic context, which comprises features for which processing is optional. Examples of intrinsic context might be the color or font in which words are printed during presentation to the subject, whereas the environmental characteristics of the room in which testing takes place would be an example of extrinsic context. Having adopted the extrinsic–intrinsic view (Godden & Baddeley, 1980), my colleague and I subsequently abandoned it as being too passive and stimulus determined. The concept of interactive context places the emphasis on the processing carried out by the subject rather than on the characteristics of stimulus material. Hence, as Gardiner and Tulving (1980) showed, when subjects are given the task of learning numbers with abstract words as context, the two features of the stimulus configuration tend to be processed independently. However, if subjects are given plenty of time and are instructed to integrate the two elements into a single item, then they tend to process the context interactively.

As in the case of the concept of processing domains, the concept of interactive versus independent context cannot be tested by means of a single experiment. It should be evaluated first on the basis of whether it is useful in giving a convincing account of existing data and, second, on the question of whether it generates research leading to a better understanding of the role of context in memory.

Recollection

The term *recollection* is used to refer to the active problem-solving aspect of retrieval that subjectively appears to play an important part in remembering. Although such factors have sometimes been discussed (e.g., Lindsay & Norman, 1972, p. 379), they have largely been ignored in laboratory studies of retrieval during the 1970s. In some ways this is surprising because the active role of the subject in organizing and learning formed an important tenet of cognitive psychology during this period. It is interesting to note that Tulving (1962, 1966), one of the strongest advocates of the view that learning depends on active organization by the subject, has emphasized an essentially passive interpretation of retrieval. Underpinning the concept of encoding specificity (Tulving & Thomson,

1973) is the concept of a set of retrieval cues that automatically evokes the appropriate response. The concept of recollection does not deny that such automatic processes form an important component of retrieval. It suggests, however, that they offer only a partial account of the complex process of retrieval. We are able to "search" our memory probably by actively setting up plausible retrieval cues. More importantly, when trying to remember something, we do not blindly accept every candidate memory as veridical. We have ways of evaluating memories and subsequently using these to refine our recall of an event or situation. The term recollection is used to refer to this active process of setting up prospective retrieval cues, evaluating the outcome, and systematically working toward a representation of a past experience that we find acceptable. Labeling this set of processes "recollection" does not of course constitute an explanation; it is, however, a reminder of an important aspect of human memory that has been sadly neglected.

Before going on to illustrate the development and application of these concepts, it is perhaps worth discussing briefly at this point the issue of their originality. They are original in the sense that they represent an attempt to come up with ideas that would help us understand data that I believe could not be adequately represented by currently available concepts. They are unoriginal in two senses. First, like any new concept, they derive from already existing ideas. Hence, the concept of a domain is a modification of levels of processing, whereas both it and levels of processing owe something to the concept of a domain presented by Sutherland in his theory of the visual perception of objects (Sutherland, 1973). It does, however, differ from both these concepts in certain important ways and in that respect is original.

A second sense in which many of the ideas that follow are not original stems from the fact that other investigators were independently arriving at broadly similar concepts at the same time. In some areas these probably came at least in part as a result of a common attempt to account for similar data. The conceptualization of the retrieval deficit in amnesia is a case in point (Baddeley, 1982; Weiskrantz, 1978; Baddeley & Brooks, Note 2). Other cases occur where parallel results were being obtained in two related but rather different areas, as in the case of context dependency. This was studied in divers by Godden and myself (Godden & Baddeley, 1975, 1980); under more standard laboratory conditions by Smith, Glenberg, and Bjork (1978); and via the condition of state dependency by Eich (1980). In these cases the phenomena observed were original in the sense that any one study was not directly influenced by the others but unoriginal in the sense that equivalent phenomena were observed by all three groups of investigators.

Some Applied Problems

Context-Dependent Memory in Divers

It has been known for many years that material learned in one environment may often be recalled better in that environment than in an alternative setting. Although it is sometimes difficult to produce a reliable effect in the laboratory (e.g., Strand, 1970), my colleagues and I incidentally observed what appeared to be a massive effect of context dependency when divers were asked to recall passages of prose or lists of unrelated words learned under water (Baddeley, Cuccaro, Egstrom, Weltman, & Willis, 1975; Davis, Baddeley, & Hancock, 1975), though neither study gave any indication of context dependency when memory was tested by recognition. In an experiment designed specifically to investigate context dependency, Godden and I (Godden & Baddeley, 1975) had divers learn lists of words either on the beach or under about 15 feet (4.55 m) of water. When the recall environment differed from that in which learning took place, there was a decrement of over 30% in number of words remembered.

Such a result has obvious practical implications; it suggests that diver training should not rely too heavily on teaching on dry land and, furthermore, indicates that if divers are being used as inspectors, then they should record their observations as they make them and not rely on memory. From a theoretical point of view, this result seems to fit very neatly into the framework suggested by the encoding-specificity hypothesis: Words that are originally encoded in the context of the underwater environment will be most easily recalled when that context is revived.

In a subsequent experiment (Godden & Baddeley, 1980), we extended our study to look at the influence of environmental context on recognition memory. Once again we used divers who were required to learn lists of words either on land or under water. This time, however, they were tested by recognition instead of recall. We obtained a small decrement when learning took place under water but no suggestion of a context-dependency effect, with subjects recognizing 72.0% of the words when the environment was reinstated and 72.5% when it was changed.

A search of the literature reveals other instances where changing the environmental context appears to influence recall but not recognition (Smith et al., 1978). A similar pattern of results has also been obtained in the case of state dependency, where the subjects' internal environment is changed by their ingestion of a drug, such as alcohol. A recent extensive review of the literature on state-dependent memory by Eich (1980) demonstrates very convincingly that the effect is crucially dependent on cueing conditions and is virtually never observed when memory is tested by recognition. Such findings are in clear contrast to the many demonstrations that changing the verbal or semantic environment within which a word is learned may have a very dramatic effect on both its recallability and recognizability (Light & Carter-Sobell, 1970; Tulving & Osler, 1968).

The encoding-specificity principle has of course no difficulty in explaining why context influences recall; if the environmental cues were encoded during list learning, then there is every reason for them to serve subsequently as retrieval cues, enhancing recall. It is, however, far from obvious how one can account for the failure of environmental context to influence recognition. It is clear from our first study that divers do encode context, but equally clear from our second that this does not influence their recognition performance. Because recognition involves retrieval (Tulving & Thomson, 1973), why does the environment not provide a retrieval cue? A contextual-tagging theory (e.g., Anderson & Bower, 1973) is no more successful in resolving this paradox. If the environment makes the contextual tags more accessible or discriminable in the recall condition, then it should be equally helpful in recognition.

Our results seem to demand two modifications to the view of retrieval suggested by the early version of encoding specificity. First, Tulving and Thomson (1973) state clearly that they "make no theoretical distinction between recall and recognition" (p. 354). Our results imply a clear distinction. Second, our results suggest that Tulving and Thomson's rejection of two-stage generate-and-recognize models of retrieval may have been premature. Although they were no doubt justified in arguing against certain specific two-stage tagging models, our data indicate that environmental context will influence the accessibility of the memory trace (generation) but will not influence the subject's ability to evaluate the trace as that of a "new" or "old" item (recognition). Our diving data are particularly cogent here because the dramatic environmental change induced by diving produces an extremely large contextual effect, making it quite clear that the absence of such an effect under conditions of recognition is not simply due to the insensitivity of the experiment. Our data then argue for a distinction between recall and recognition and favor a two-stage model.

A problem raised by our result stems from the contrast between environmental context, which we found did not influence recognition, and verbal context, which has been shown by Tulving and Thomson (1973) and many others to be a potent factor in recognition. The proposed distinction between interactive and independent context offers a plausible explanation for this apparent discrepancy. When a verbal stimulus word is accompanied by a low-frequency associate, the two are likely to interact, hence influencing the subject's encoding and interpretation of the TBR item. In the case of environmental context, however, there is no reason why the interpretation of a randomly selected set of words should depend on whether they are experienced on land or under water. The environmental context will therefore be encoded independently and in parallel. Such an encoding may enhance the accessibility of the memory trace laid down during learning. Under conditions of recognition, however, access to the trace is not a major problem; the difficulty comes in deciding whether the word represented was in fact previously presented. This evaluation is clearly not influenced by independent context.

Face Recognition

During the 1970s there was a considerable growth of interest in memory for faces. This stemmed very largely from a concern with the role of the eyewitness in legal cases, and much of the work has been concerned with demonstrating the unreliability of witnesses and the ease with which eyewitness testimony may be manipulated (Clifford & Bull, 1978; Loftus, 1979). Our own interest was slightly different in that we were concerned with the problem of whether one could improve a person's ability to remember and recognize faces.

We first studied a training course that was designed on the basis of the view that performance could be improved by teaching the trainee to analyze a face into its constituent features and then categorize these, an approach advocated by Leonardo da Vinci (Gombrich, 1962, p. 294) and more recently popularized by Penry (1971). Unfortunately, the only effects of this course were to make the trainees perform more poorly under certain recognition conditions (Woodhead, Baddeley, & Simmonds, 1979).

A more promising approach to the problem seemed to be offered by levels of processing, which suggests that faces might be remembered better if they are processed more deeply. A study by Bower and Karlin (1974) claimed to have demonstrated this. Subjects were given either the "shallow" task of deciding whether each of a sequence of faces was male or female or the "deeper" task of assessing the intelligence or honesty of the people represented. The deeper encoding led to better subsequent recognition. However, it is quite possible that judgments of sex might be made much more rapidly than judgments of honesty or intelligence so that subjects in the shallow condition simply spent less time observing and processing the faces. If so, this result could simply be another demonstration of the total-time hypothesis, whereby the greater the amount of time spent in learning, the better the subsequent recall or recognition (Cooper & Pantle, 1967). We therefore decided to conduct a stricter test of the levels-of-processing interpretation of face recognition.

In our study (Patterson & Baddeley, 1977), subjects categorized faces on the basis of either physical dimensions, such as nose size or thickness of lips, or personality dimensions, such as nice–nasty and intelligent–dull. Our subjects were subsequently required to recognize the faces among a range of distractors. We observed a significant but disappointingly small effect of encoding instructions. At the same time, we observed massive effects of disguise and substantial effects of pose, but neither of these interacted with encoding strategy. Our failure to find a major effect of encoding is consistent with other studies that suggest that, provided the subject is required to treat the face as a whole rather than as a set of independent features, the manner of encoding is of little significance. Hence, clearly semantic judgments such as assessment of friendliness are no more effective than apparently shallow judgments such as how heavy the subject is (Winograd, 1976). Such findings suggest that the important factor is not depth of processing but rather whether the subject processes the face as a set of isolated features or as a single gestalt in which the whole range of physical features participate and interact.

By this stage in the project, however, the concept of levels of processing had been modified by Craik and Tulving (1975) to include the concept of elaboration. It seemed possible, therefore, that if a genuine depth-of-processing effect was occurring, then it might be possible to enhance it by inducing a greater degree of semantic elaboration. We attempted to achieve this in a series of studies in which faces were presented either together with simply a name and age under control conditions or accompanied by a semantic context, describing the person's occupation, background, and habits (Baddeley & Woodhead, 1982). We conducted two such experiments. The first showed a nonsignificant advantage for the elaboration condition

(66.5% correct; $d' = 2.18$) over the control (61.5%; $d' = 2.03$), whereas the second showed a nonsignificant tendency in the opposite direction, with the elaborative condition (73.5%; $d' = 1.83$) being marginally worse than the two control conditions used (75.5% and 72.2%; $d' = 1.98$ and 2.08, respectively). Neither experiment, therefore, provided any evidence for the suggestion that semantic elaboration of faces would enhance subsequent recognition.

By this time we were becoming a little disenchanted with a levels-of-processing approach to face recognition. Not only had we failed to amplify our previous small effect but we had lost it altogether. Why should that be? A possible interpretation is suggested by Fisher and Craik (1980), whose study using verbal material indicated that although elaboration may enhance recall, it will only influence recognition when the elaborating material is highly compatible with the material to be remembered. It seems likely that our random assignment of contexts to faces would not produce a particularly compatible mapping. Our results are therefore consistent with those of Fisher and Craik, who themselves suggest that their results should generalize to faces. Our views concur with those of Fisher and Craik in suggesting that these results argue against the original Craik and Tulving (1975) position that elaboration has its effect by producing an enriched and hence more discriminable trace of the target item. Such a view predicts a clear effect of elaboration on both recall and recognition. Our data suggest that this will occur only if the item and its context are encoded interactively, a type of coding that is likely to be difficult unless target and context are reasonably compatible.

Fisher and Craik (1980), however, suggest that even incompatible context should be effective if it is reinstated at the time of test: "Elaborative encoding will be effective to the extent that the encoding context is reinstated by the retrieval context" (p. 404). Some evidence for this review is produced by the studies of Watkins, Ho, and Tulving (1976) and Thomson (in press), all of whom observed contextual effects in face recognition. We attempted, therefore, to test this prediction by manipulating the presence of elaborative information during both the learning and recognition of faces.

Table 1
Effect of Context on Face Recognition

Measure	Context: Correct	Mismatched	New	Absent
Hit rate[a]	76	69	54	58
False-alarm rate[a]	45	45	24	27
d'	.78	.61	.75	.76
$\log \beta$	−.23	−.14	.17	.15

[a] Figures are percentages.

We therefore presented 24 faces, all of which were accompanied by a contextual statement such as "cook in a hamburger bar," "diver on an oil rig," or "barman in a village pub." The faces were then mixed in with an equal number of new faces and tested under one of four conditions: with no context present, with the appropriate context, with a context that had previously been presented with another face, or with a new and, hence, unfamiliar contextual statement. Table 1 shows the results.

In this study we clearly did get effects of context, with targets being recognized 76% of the time when accompanied by the appropriate context as opposed to 58% when context was absent. Note, however, that subjects were not significantly less likely to detect a target accompanied by a familiar but mismatched context (69%). If one looks at the false-alarm responses, a similar biasing effect occurs, with subjects being reliably more likely to respond yes to a new item accompanied by an old context. Finally, by analogy with the results from verbal memory, one might expect that presenting an unfamiliar new contextual label would inhibit the correct recognition of old items but enhance the probability of correctly rejecting new items. As Table 1 shows, this does not occur; a totally new context leads to a level of performance equal to that found with no context.

This pattern of results becomes clear if both hit and false-alarm rates are combined to separate detection sensitivity (d') and bias ($\log \beta$). Context has no reliable effect on sensitivity but does affect bias; familiar context increases the probability of responding old to both "new" and "old" faces, in contrast to a new or absent context.

Overall, then, our results appear to suggest that the attempt to apply the concept of levels of processing to recognition of faces was premature. At a practical level, at least, it seems unlikely that inducing the subject to encode a face deeply or elaborately will necessarily enhance subsequent recognition. However, although the concept of levels of processing has not proved helpful in studying face recognition, I would like to suggest that the concept of a domain, and a subdomain, does remain a useful one.

Although the concept of a domain does not of itself generate testable predictions, it does provide a framework for discussing face recognition. Consider first the learning and recognition of unfamiliar faces. It is likely that the face is processed as a gestalt rather than being processed as a set of independent additive features. The combination of these features gives rise to the perception of a person who often will be seen as having certain characteristics, for example, ugly or handsome, intelligent or stupid, shifty or reliable, Nordic or Arabic. One might therefore distinguish two subdomains, one concerned with features and the topography of faces and the other concerned with their semantic associates, real or imagined. The existence of stereotypes indicates reasonably strongly the connections between these two subdomains.

Consider now the face of someone who is familiar, for example, ex-president Nixon. This conception will not only include topographic characteristics together with one's assessment of such characteristics as whether he looks trustworthy but will also be associated with a great deal of semantic information peculiar to this particular face: the fact that he was president, the association with Watergate, McCarthyism, and so forth. In this case we have a very clear link with a much broader domain of our memory system than is the case with an unfamiliar face. Similarly, the face of a member of one's family clearly conjures up enormously rich associations that go well beyond the domain of physiognomy.

A second issue raised by our face-recognition results concerns the role of context. It is apparent that under certain conditions, contextual cues such as the clothes worn by the person to be recognized or the environment in which he or she was initially seen may have a dramatic effect on recognition (Thomson, in press). Although these results may be principally due to response-bias effects, there is at least one study by Watkins et al. (1976) that avoids this problem by using a two-alternative forced-choice procedure, and which nevertheless does obtain effects of context on subsequent recognition. Why the discrepancy with our own results? I would suggest that the crucial factor is the activity of the subject; if the subject is able to relate the context and stimulus in an interactive way, then context will influence both recall and recognition performance. It seems likely that our random assignment of face and context without specific interactive instructions was insufficient to produce interactive encoding. Watkins et al. went to rather greater lengths than we did to ensure such encoding. In the case of Thomson's (in press) experiments, I assume that the perception of people acting in an environment may well be affected both by their clothing and their actions. A subject shown a picture of a man in jeans and T-shirt fleeing from a bank may well interpret and perceive the target quite differently from the way he or she perceives the same person dressed in a suit and standing in a church.

What will determine whether context functions in an interactive or independent way? Interactive contextual effects seem most likely to occur within the same domain, where rich prior associations allow the item and its context to interact easily to create a novel episode. Low-frequency associates to target words seem particularly likely to present this set of characteristics, and they have, of course, been used with great success in many experimental studies. In an unpublished verbal-memory study by Muriel Woodhead and myself, we used animal names and adjectives, assigning the particular adjective to the particular animal at random (Woodhead & Baddeley, Note 3). We found no evidence of cue-dependent remembering in a subsequent recognition test, presumably because pairs like *rabbit–large* and *stoat–speedy* did not give rise to episodes that were very different from those that would have been created by the pairs *rabbit–speedy* and *stoat–large*. One suspects then that for good interactive contextual effects to occur, the material needs to be sufficiently compatible to allow a new integrated episode to be created but not so compatible as simply to reproduce a cliché.

Gardiner and Tulving (1980) discuss a related phenomenon in exploring the failure to recognize recallable words. They presented their subjects with abstract words together with retrieval cues comprising other abstract words (e.g., *moment* ABILITY or *27* BEING). Under these conditions the standard retrieval-cuing advantage did not occur. However, a second study showed that when given plenty of time and a clear instruction to integrate the elements of each pair into a single item, results approximated much more closely the normal retrieval-cuing function. When stimulus and context come from separate domains or subdomains with few associative links, then the context will contribute a purely independent effect unless the subject can integrate stimulus and context into a new and unitary episode. Gardiner and Tulving's result therefore seems related to our own attempts to enrich encoding of faces by providing semantic context involving a purely arbitrary mapping of a semantic domain onto a facial domain. The results showed that subjects were retaining the information about both but that the link between the two failed to be established. In this situation the context remained independent, causing no restructuring of the facial information. Although it seems likely that both item–context compatibility and the subject's strategy will influence the likelihood of contextual effects becoming interactive rather than independent, we clearly need considerably more empirical evidence on this issue.

Amnesia

A series of collaborative studies were aimed toward applying the concepts of normal memory to the study of amnesic patients. The first of these (Baddeley & Warrington, 1970) explored the hypothesis that amnesic patients have normal short-term but defective long-term memory; despite some interesting discrepancies from the straightforward picture, the hypothesis was broadly supported. We went on to explore the possibility that the amnesic deficit might be a result of defective semantic coding and found that in a verbal free-recall task, our amnesics were able to take as much advantage as normal subjects of phonological cues, were somewhat defective in using taxonomic category cues, but showed no evidence of benefit from a visual-imagery mnemonic (Baddeley & Warrington, 1973). Such a pattern of results could be interpreted broadly in terms of a levels-of-processing explanation of amnesia.

A levels-of-processing view was also vigorously advocated by Cermak and his associates following a quite separate and independent series of studies in which amnesic alcoholic Korsakoff patients were shown to gain a normal advantage from rhyme but not semantic cues (Cermak & Butters, 1972; Cermak, Butters, & Gerrein, 1973). Perhaps the most dramatic demonstration, however, was that provided by Cermak, Butters and Moreines (1974) using the release-from-proactive-inhibition (PI) technique, indicating that their amnesic patients showed normal release effects when material was switched from letters to digits but showed no release when word sequences were used and taxonomic category switched.

Levels of processing therefore appeared to provide a promising interpretation of the dramatic memory loss found in amnesic patients; it was suggested that amnesics have memory problems because they simply do not spontaneously encode material deeply enough. This possibility was explored by three separate groups of investigators: Cermak and his colleagues in Boston; Mayes and Meudell in Manchester, England; and Neil Brooks and myself, using primarily Glasgow patients. We all tested the hypothesis that amnesia might be due to an encoding deficit by ensuring that our amnesic and control patients encoded the material deeply and richly and by then testing recall.

If the amnesic deficit is due to the poor encoding strategies spontaneously adopted by amnesics, then requiring deep encoding should dramatically improve their performance. If the problem stems not from the strategy adopted but from a failure to be sensitive to level of encoding, one should find no effect of manipulating level of processing. What emerged from all three research groups was the finding that provided ceiling and floor effects were avoided, amnesics responded to level of encoding in just the same way as did normal controls. Their overall level of performance was, however, dramatically lower. We all, therefore, came to the same conclusion, namely, that ensuring that amnesic patients processed material deeply did not substantially alleviate their memory problems (Baddeley, 1982; Cermak & Reale, 1978; Mayes, Meudell, & Neary, 1978). It has furthermore become clear that although the Korsakoff patients studied by Cermak do indeed appear to show a rather narrow and shallow range of encoding strategies, this is by no means true of all amnesics (see Baddeley, 1982; for a discussion of this). Conversely, there appear to be cases of patients with frontal lobe damage who also show a failure to release from PI when taxonomic category is shifted but who are not grossly amnesic (Moscovich, 1982).

It is clear, then, that levels of processing does not offer the elegant interpretation of amnesia that at first seemed likely. It is of course unfair to criticize levels of processing on this basis because the concept was not initially devised as an interpretation of amnesia. However, the fact that massive differences in memory performance do exist between amnesics and control patients, and that these do not conform to the predictions of levels of processing, implies at least other major factors in human memory. A clue as to what these factors might be is given by the examination of memory tasks that are not impaired in amnesic patients.

In a classic observation, Claparède (1911) reports an incident in which he concealed a pin in his hand when shaking hands with an amnesic patient. On the following day, the patient was unwilling to shake hands but was quite unable to offer any explanation for his reluctance. Subsequent research has shown that amnesics are capable of a wide range of tasks, including classical conditioning, the acquisition of motor skills, learning to assemble jigsaw puzzles, learning to perceive Mooney (1957) figures, or learning to detect anomalies in cartoons (Baddeley, 1982; Meudell & Mayes, 1982). In all these cases patients show unmistakable evidence of learning but no indication of conscious memory of the task on which they show learning. The patient appears to show evidence of unconscious learning but not of conscious recollection.

One of the neatest of many recent demonstrations of this phenomenon comes from Jacoby and Witherspoon's (1982) use of a number of homophones (e.g., *read* and *reed*). The less common word of the pair was incorporated in a definition, and the subject was invited to provide that word (e.g., Question: "What is the part of the clarinet that vibrates?" Answer: "The reed"). After giving a number of such definitions, the experimenters asked the subject to recognize the words they had generated from a mixed list of new and old words. Amnesic patients performed extremely poorly on this task. Subjects were then asked to spell a number of words, no mention being made of prior learning. There was a clear tendency for the spelling to reflect the form of the homophone that had previously been defined (e.g., *reed* rather than *read*). This tendency was just as strong in amnesics as in controls, showing clear evidence of some form of learning, despite the inability of the patient to recall or recognize the words spelled.

What do these various tasks that are spared have in common? A suggestion made by Weiskrantz (1978) and myself (Baddeley, 1982; Baddeley and Brooks, Note 2) is that in all these cases the subject's memory is tested using procedures that do not involve conscious recall. Retention of a particular item or incident is tested indirectly by asking the patient to perform a task on which performance is to some extent dependent on retention of earlier information. In a tracking task, for example, the subject simply attempts to perform the skill to the best of his or her ability; the advantage of having practiced previously does not depend on consciously remembering the previous practice session, and indeed, the amnesic patient will typically claim not to have encountered the task before. Similarly, in perceptual tasks, the patient does not need, for instance, to remember having seen a Mooney figure before; he or she simply needs to attempt to make sense of the pattern of black and white patches. It seems unlikely that conditioning will depend on actively recalling a specific prior event, while Jacoby and Witherspoon's subjects do not need to be aware of the bias produced by the prior definition task in order to be influenced by it.

Weiskrantz (1978) draws the interesting analogy between amnesic patients and "blind sight" in which certain patients with visuo-cortical defects are unable to report conscious experience of material presented to parts of their visual field but when forced to "guess" perform correctly. The necessary information is available, but they are not aware of it. In the case of amnesic patients, it appears that they do learn and can acquire new material, but they are not aware of this and consequently cannot perform tasks that depend on such awareness. According to this interpretation, then, there are two separable aspects of learning, a relatively automatic process that shows little or no disruption in amnesic patients and a more active retrieval process based on conscious awareness. The process of recollection will depend on this conscious awareness of familiarity to guide both its search and evaluation processes.

How is the concept of recollection related to the encoding-specificity hypothesis? Tulving and Thomson (1973) themselves point out that their hypothesis is only a partial answer to the interpretation of retrieval: "A modest beginning has been made, and it is the purpose of the present paper to examine and describe the early state of the art" (p. 353). There is no doubt that conscious retrieval, at least, does have a substantial problem-solving component that operates at two levels, one being concerned with the attempt to generate plausible candidates for the target word and the other being concerned with checking these out and deciding whether they are correct. Such a view is, of course, far from novel (e.g., Bartlett, 1932; J. Brown, 1976; Jacoby & Craik, 1979).

Solso (1979, p. 208), for example, refers to two protocols from students asked to recall the names of their fourth-grade teachers. One protocol described first of all recalling which school was attended during fourth grade, then locating the classroom and then visualizing the teacher—"tall and thin and the same teacher who had taught during third grade." This produced the hypothesis "Miss Bell?" which was followed by apparently irrelevant information as to which teachers she was friendly with, when the particular school was entered, and where the classroom was, followed by a convinced response that it was indeed Miss Bell.

In both this and other recollections (Baddeley, 1982), there is a stage of attempting to provide as much relevant context as possible, followed, if successful, by the "popping up" of a response, which is then checked out. The initial "search" process of generating associated information seems to be a conscious strategy; when it works it is plausible to assume that it has done so because it has led to the generation of an effective retrieval cue. The encoding-specificity hypothesis is concerned with the operation of such retrieval cues. The concept of recollection incorporates this as an important component but adds the active process of generating associated information in the hope of encountering a retrieval cue, as well as the process of evaluating any candidate responses that may be evoked.

The evaluative process is a very important but little understood one. In earlier models of recognition, the problem tends to have been finessed by simply referring to finding list or temporal tags attached to nodes in semantic memory. Such analogies probably have more to do with the techniques of 1960s computer science than to the way in which human memory works. However, we still know very little about what it is that convinces us that we know something or that we have recalled a name or an incident correctly.

It seems probable that there is a range of sources of evidence of correct recall. One such source appears to involve generating further information from the item retrieved that can then be used to check out the plausibility of the response. In Solso's (1979) example, such apparently irrelevant detail as the location of the classroom and who were Miss Bell's friends apparently served this function and led to an increased certainty about the correctness of the name recalled.

Another source of evidence as to the correctness of a recall is probably one's general knowledge of the world. In the second protocol given by Solso (1979, p. 208), a student trying to remember the name of his teacher recalls that her name began with an *A*, then *Al*, then produced *Alvira*. He then remembered that her name was similar to a province in Canada and hence rejected *Alvira* in favor of *Alberta*.

A further source of evidence that a response is correct stems from one's own memory. It is possible, for example, that a subject may use speed of response as an indication of correctness. For example, Mandler (1980) postulates two processes in recognition, one based on retrieving relevant context, a process that allows the learning episode to be recalled, and the other based on judgment of familiarity. Mandler suggests that one possible cue to the fact that an item has been previously experienced is the facility with which it is processed. Such a cue is occasionally used by amnesics. In a task involving the reordering of the words in scrambled sentences, we asked an amnesic patient who had completed a sentence on several occasions previously if he had seen that sentence before. He replied that it did not seem at all familiar, but because he had done it so rapidly, he assumed that he must have done it before.

Another piece of evidence that is probably used is that of "had this been the case, I would have remembered." For example, walking to a meeting in a building I had visited two or three times before, but that was in an area of London with which I am not particularly familiar, I suddenly had the feeling that I must have taken the wrong route because I saw a large statue of a dragon inscribed, *City of Westminster*. I assumed that had I been that way before I would surely have noticed it and remembered it. In fact, it proved to be the case that I was on the correct route but in the past had looked the other way and, hence, had not noticed the dragon. J. Brown, Lewis, & Monk (1977) have investigated this factor in verbal-recognition memory.

It is, of course, frequently the case that we know we are correct with no apparent need to use such complex and indirect checking devices. If asked your name, you are unlikely to seek very much confirmatory evidence before responding with a fair degree of confidence. On what do you base your confidence? One can only speculate, but it may be that it is based on an implicit comparison between the strength of response made and the strength of any competing responses to the same question; if there is a clear rapid answer and there are no competitors, the system might plausibly assume that the response is correct.

Everyday memory may clearly involve many such confident responses; it may also, however, frequently involve a good deal of conscious evaluation of material retrieved. As we have seen, amnesic patients appear to be defective in this evaluative stage, even when they can generate the appropriate item. Conversely, aphasic patients are often unable to generate the appropriate name but can recognize it as correct once it is provided. Regrettably, we know virtually nothing of this evaluative component of human memory, despite the plea that concluded Tulving and Madigan's (1970) influential review of memory written some 10 years ago:

> What is the solution to the problem of lack of genuine progress in understanding memory? It is not for us to say, because we do not know. But one possibility does suggest itself: why not start looking for ways of experimentally studying, and incorporating into theories and models of memory, one of the truly unique characteristics of human memory: its knowledge of its own knowledge. (p. 477).

Why not indeed?

To what extent is the occurrence of learning in the absence of conscious remembering peculiar to amnesia? Meudell and Mayes (1981) studied this question using both normal and amnesic patients who were required to search cartoons for particular objects. Over repeated trials, both groups showed clear evidence of learning as evidenced by faster location of the objects. When tested after a delay of 7 weeks, the amnesics continued to show speedy performance, indicating clear evidence of retention, but were quite unable to recognize the relevant cartoons—the classical dissociation between learning and awareness. However, the controls showed an exactly equivalent effect when tested after a much longer delay of 17 months, indicating that the phenomenon is not peculiar to amnesia but is probably characteristic of weak memory in general.

As Jacoby and Witherspoon (1982) point out, the Claparède phenomenon has other counterparts in the study of normal memory. For example, Kolers (1976) required his subjects to read sentences presented in transformed text. He found that having read a specific sentence in a particular transformation led to faster reading of that sentence even after a year's delay, indicating very durable and very specific learning. In contrast, recognition memory was relatively poor, and the correlation between memory measured in terms of speed and recognition was very low.

Jacoby and Dallas (1981) have further explored the distinction between these two types of memory. They term one *autobiographical memory* and the other *perceptual memory*. In their studies, autobiographical memory is measured by recognition, whereas perceptual memory is measured by a task in which words are presented tachistoscopically and evidence of prior learning provided by a greater probability of detection at a standard brief exposure. In one study, Jacoby and Dallas varied the depth of processing of the words presented, subsequently testing retention by either recognition memory or visual detection. Both techniques showed evidence of learning. However, whereas recognition memory produced clear evidence of the standard levels-of-processing effect, with semantic encoding leading to better performance than phonological or orthographic processing, level of encoding had absolutely no effect on memory as measured by detection threshold.

Although this result again argues for at least two separate types of learning, the nature of the process Jacoby and Dallas term *perceptual learning* is still far from clear. One possibility, which Jacoby and Dallas argue against, is that some form of semantic memory is involved. Morton (1979) argued for a priming effect within the visual-input logogen system. Such an interpretation might suggest that other priming effects could occur in other processing domains, and data from amnesics does indeed suggest that this type of learning occurs across a much wider range of tasks than could reasonably be regarded as constituting perceptual memory. Examples include the acquisition of motor skills, conditioning, completing jigsaw puzzles, and unscrambling scrambled sentences (Baddeley, 1982). The nature of the two types of learning suggested both by data from amnesic patients and by the work of Jacoby and Dallas presents an important, exciting question.

Retrieval and Aphasia

The final area of practical importance discussed is one that has not traditionally concerned memory researchers, namely, the understanding of word finding in aphasic patients. It is commonly the case that damage to the left side of the brain produces substantial problems in the normal utilization of speech, often associated with a particular difficulty in retrieving the names of objects. One of the roles of a speech therapist is to try to help such patients recover their ability to retrieve words efficiently and appropriately. There is, however, relatively little understanding of this retrieval process. The section that follows represents an attempt to take the concepts of retrieval developed in the study of episodic memory and attempt to use these to throw light on the question of word finding both in aphasic patients and in normal subjects.

Once again, the concepts of levels of processing and encoding specificity will be used. It is, however, important to emphasize that neither of these concepts was explicitly designed for application to semantic memory. Although that does not prevent them from being used as a source of ideas, any failure of the two concepts to explain problems of word finding in aphasia does not of course reflect on the usefulness of the two concepts in the area of episodic memory for which they were originally devised. Whether they are successful, the attempt to apply such concepts to word finding should have interesting implications for similarities and differences between semantic and episodic memory.

It has often been suggested (e.g., Howes & Geschwind, Note 4) that aphasic patients function basically like normal subjects except that their processes of word retrieval are slowed down. If this is the case, it implies both that normal concepts of word retrieval may help one understand aphasia and that studies of aphasia may help our understanding of normal word retrieval. With this in mind, Hatfield, Howard, Barber, Jones, and Morton (1977) studied the influence of context on object naming in aphasic patients.

An appropriate context has been shown to facilitate the perception of both words (Morton, 1969) and objects (Palmer, 1975). Furthermore, speech therapists have claimed that aphasic patients find it easier to access the name of an object in the context of its use, for then "the patient will be led into the mental attitude in which the words occur. This makes it much easier for him to retrieve the appropriate word" (Hatfield, 1971, p. 24). A similar view was put forward by Goldstein (1948), who claims that "some words, which occur easily in fluent speech, offer particular difficulty in voluntary word finding. . . . The patient cannot find the words because he cannot assume the attitude in which they normally appear", (p. 60). Such a view would, of course, be quite compatible with an encoding-specificity interpretation; the name of an object is likely to be acquired in the context of that object and its utilization, and hence, a realistic representation of the object in an appropriate context should maximize the probability of recalling its name.

However, although it was generally believed that an appropriate context and a realistic representation helped aphasics in word retrieval, the available evidence was far from convincing. Although Bisiach (1966) found that his patients had more difficulty in naming outline drawings of objects than they did "realistic coloured figures," Rochford and Williams (1962, 1965) found little evidence for the importance of realism in object naming. In one such study, Rochford and Williams (1962) had their subjects name objects such as the handle of an umbrella. If the aphasic patient failed, he or she was then cued in one of four different ways: (a) a functional description of the object was given, (b) a strongly constraining sentence-completion cue was provided, (c) a rhyming word was given, or (d) the appropriate name was spelled out. Cues b, c, and d were all equally effective, whereas a functional description, though effective, was only half as useful.

In a direct test of the effect of realism, Rochford and Williams (1965) required their patients to name either eight pictures of objects or eight parts of the experimenter's body that were pointed to. There was no difference in probability of naming between the pictorial representation and the real body parts. Although there are clearly problems in interpreting this last result due to the nonequivalence of the two types of items, it does cast doubt, nevertheless, on the view that greater realism enhances object naming.

The observation that a rhyming word is a better cue to the name of an object than a functional description would appear to be at odds with any suggested interpretation in terms of either levels of processing or encoding specificity. It creates problems for a simple levels-of-processing interpretation because it seems to suggest that phonemic rather than semantic encoding is more useful, whereas a simple encoding-specificity hypothesis would be faced with the problem of explaining why a rhyming word (e.g., *candle*) should serve as a good retrieval cue for *handle* when the two must rarely, if ever, have been presented to the subject together. In view of the importance of this point and the comparative lack of evidence to support it, Hatfield et al. (1977) decided to investigate in more detail the role of realism and context in word finding by aphasic patients.

Hatfield et al. performed three experiments. In the first, drawings of objects in different situations were used. One condition consisted of the object in a highly plausible context: For example, the object might be a brush with a drawing of a girl looking under a bed for the brush. This was accompanied by the sentence "Ann is looking under the bed for the ____" A second condition involved a somewhat less plausible scenario, "Albert hits the nail with a ____," whereas the third represented the least plausible condition, "The snake is hiding in the ____"; all were illustrated by appropriate drawings. Although the probability of naming an object did increase over three successive presentations, there was no evidence that appropriateness of the context enhanced naming.

In their second study, Hatfield et al. (1977) manipulated visual rather than semantic context, drawing a series of objects either alone or in an appropriate context. An example might be that of a snail, pictured together with a drawing of flowers, leaves, grass, and so forth. In this study the probability of recalling the items when presented free of context was .56, whereas in a context it was .55, once again indicating no facilitation of naming by context.

The final experiment by Hatfield et al. was a direct test of the role of realism in naming. They presented their aphasic patients with objects, photographs of such objects, or line drawings and compared the probability of the subject producing the correct name. There was no significant difference between conditions, with naming probabilities for the objects, photographs, and drawings being .84, .76, and .77, respectively.

In accounting for their results, Hatfield et al. distinguished three stages in their naming task: recognition of the objects, leading to recovery of knowledge about the object that is then used to find the name. They suggest that the role of context in tachistoscopic recognition of words and objects by normal subjects (Morton, 1969; Palmer, 1975) occurs because under such degraded stimulus conditions the first stage, namely, that of object recognition, is the limiting factor on performance. In the case of aphasic patients, they have no difficulty in perceiving the object and knowing what its purpose is; they simply cannot access its name. That being so, improving the realism of the stimulus is unlikely to be helpful because it merely enhances a stage that is already functioning perfectly adequately. What is required is assistance in accessing the name. This can be facilitated in the short term, at least, by priming within the lexical domain; hence, a patient is able to repeat a word he or she has just heard. Hatfield et al. (1977) found evidence of priming in all three studies because the probability of retrieving an item increased over successive tests; it appears that having retrieved the name once, the patient has for a short time a higher probability of retrieving it, even though the retrieval context is dramatically changed (e.g., from a brush with a snake in it to a girl looking for a brush under her bed).

A second way of enhancing lexical retrieval is by spelling the word out or using a rhyme (Rochford & Williams, 1962). This presumably works because associations within the lexicon avoid the need to access the name code directly from the semantic representation. The use of cliché, for example, cuing the word *butter* with the cliché "bread and ____," is another powerful retrieval cue in this context (Rochford & Williams, 1962), once again because the associations presumably operate within the lexical domain and do not rely on a lexical–semantic link.

What then are the theoretical implications of these studies? It might seem reasonable to suppose that the name of an object such as a snail will have been initially learned primarily in the context of real snails, or at least pictures of snails. That being so, these might be expected to form more appropriate retrieval cues for the response *snail* than would a rhyming word such as *pale.* This clearly is not so. Such results are more compatible with a view that separates the various processing domains involved in picture naming. Hence, the lexical-access component of naming—the availability of the name itself—is distinguished from the visual and semantic characteristics of the task. An aphasic patient can see and know what a snail is without being able to "remember" that it is called a snail. Although such a basic distinction may seem trivially obvious, it does imply a complex relation between objects, words, and their meaning.

One might reasonably argue that these results may not apply outside the rather constrained area of aphasic speech. How plausible is it that evidence from aphasics might extend to normal word retrieval? The most obvious link is via the tip-of-the-tongue phenomenon first studied by R. Brown and McNeill (1966). They read out definitions of low-frequency words to their subjects and studied instances where subjects were certain they knew the word in question but could not produce the relevant name. Under these circumstances subjects frequently knew the number of syllables in the missing word and when urged to recall the initial letter were correct on 57% of the occasions, well above chance level. They were also able to reject plausible alternatives and subsequently respond with great confidence once the appropriate word had become available. Gruneberg (1978) required geography students to recall the names of capitals of various countries. When students could not recall a capital, they were asked to rate their strength of conviction that they did actually know it. They were then cued by being given the first letter of the name. There was a clear tendency for the first-letter cue to enhance recall, an effect that was particularly clear in the case of those capitals that were widely known by the group. In short, the first-letter cue appears to be effective but only when the information is there to be cued. The first-letter cue has also been studied in aphasic patients by both Barton (1971), who found that over 60% of retrieval failures were alleviated by the first letter, and by Goodglass, Kaplan, Weintraub, and Ackerman (1976), who found significant, though less dramatic, enhancement of word retrieval when the first letter was provided.

The importance of the first letter of a word is also demonstrated in episodic memory tested by free, paired-associate, or serial recall in a series of experiments by D. L. Nelson and his associates (Nelson, 1979). Nelson and Rowe (1969) required their subjects to learn paired-associate lists comprising three-letter words as stimuli and digits as responses. They varied the intralist stimulus similarity, having words share either initial, middle, or terminal letters. They found a clear tendency for intralist similarity to impair performance, with impairment being greatest when initial letters were shared by several stimuli.

Nelson has gone on to demonstrate across a wide range of tasks and conditions that the formal orthographic characteristics of the material play an important role in verbal learning, particularly under conditions of rapid presentation. In one study, for example, Nelson, Wheeler, and Brooks (1976) had subjects learn pairs of words varying in imageability and degree of association between stimulus and response. They found that even under conditions where imagery and associative compatibility were having a clear effect, subjects were still impaired when stimuli had similar first letters (e.g., *candle*, *cabin*). Subjects could avoid the similarity effect only when the rate of presentation was slowed down to 5 sec/per pair. Nelson's work clearly demonstrates the importance of such shallow cues as initial letter in verbal long-term memory. Such results do not fit easily into a levels-of-processing framework but are entirely consistent with a view that distinguishes between lexical and semantic domains and accepts that both are involved in verbal long-term memory.

One characteristic of the tip-of-the-tongue phenomenon is a tendency for retrieval to be dominated by a plausible but incorrect response. A little while ago I was attempting to generate the term that an optician would use for being short-sighted; the word that kept popping up was *astigmatic*, which I knew perfectly well was incorrect; yet try as I may, whenever I searched, *astigmatic* was the word that popped up, completely blocking out *myopic*, a word that I would normally not have any difficulty in producing.

A more systematic exploration of this phenomenon was carried out by A. Brown (1979), who gave his subjects definitions of relatively low-frequency objects accompanied by either a rhyming word, an unrelated word, or a semantically closely related word. He found that presenting a semantically related item actively inhibited recall relative to a totally unrelated word. Why should this occur? Presumably, if we are to talk, then it is necessary to produce a fluent string of lexically appropriate items. At many points there may be several potentially appropriate words. For the process of speech to operate smoothly, it is necessary to choose one word and inhibit the rest, and typically, it is more important to do this than to choose the exactly perfect word at any given moment. Occasionally this inhibitory process appears to fail, leading to the production of a blend between two semantically similar candidate words. Fromkin (1971) gives a number of examples of this type of speech error, including *mownly* (a blend of *mainly* and *mostly*) and *bruthle* (a blend of *bother* and *trouble*). The smooth operation of the lexicon therefore seems to require that all items synonymous with the word being produced be inhibited. Because the process of speech is fluent, the inhibition must not extend to all words.

Analogous inhibition effects have been shown in category generation by J. Brown (1968) and in free recall by Tulving and Hastie (1972) and Slamecka (1968). The simple assumption of a separate lexical domain therefore offers a framework for explaining a wide range of phenomena from naming in aphasic patients through the tip-of-the-tongue effect and initial letter cueing to the inhibitory effects of items that are semantically associated with the inaccessible name in both semantic and episodic tasks.

The nature of the semantic domain in verbal memory is a particularly crucial one. It is also an area in which there are many conflicting views and little firm agreement. I would like to suggest, however, that any concept of meaning must allow some form of mapping from words—individual discrete verbal units—onto experience of the world that must be multidimensional, continuous, and capable of registering without difficulty new experiences that have never previously occurred. As Watkins and Gardiner (1979, p. 700) point out, this means that the end result of any adequate representation of meaning cannot simply be excitation of a single node. This creates problems for an interpretation of memory that assumes tagging of specific nodes.

The problem is avoided, however, if one assumes the distinction drawn by Tulving (1972) between semantic and episodic memory. Semantic memory represents knowledge, general information that is not confined to a single experience. The meaning of words, geographical information, and facts about people or objects are all examples of information stored in semantic memory. Episodic memory by contrast is concerned with the representation of specific individual experiences. The knowledge that Borg has been the men's champion at Wimbledon for five successive years is information in my semantic memory, as is the fact that he won the 1980 championship. My recall of the particular circumstances in which I myself watched the match reflect my episodic memory. The two aspects of memory clearly often interact; my specific experience of the match was influenced by knowing Borg had won on four previous occasions, whereas my semantic memory itself was obviously modified, if only to the extent of updating the number of successive titles won by Borg.

In the case of learning word lists, the lexical representation of each word will access one or more semantic nodes that form an intermediate stage in creating episodic experiences from words. Take, for example, the experiment by Tulving and Osler (1968) in which subjects were presented with a target word (e.g., *CITY*) accompanied by a cue word that is an associate of the target (e.g., *dirty* or *village*). It is suggested that what is experienced and stored by the subject in the case of *CITY dirty* is a representation of that aspect of the city that is associated with dirt, perhaps an image of litter or smoke, possibly based on a particular remembered episode. In the case of *CITY village*, the experience is likely to be very different, possibly representing a quiet, secluded villagelike neighborhood within a city.

What will be remembered, then, is probably not primarily the lexical representation of *city*, *village*, and *dirty*, nor some simple semantic representation of each of these, but rather an episode that is created from these constituent units. A simple analogy is with that of color printing, where a small number of "pure" colors may be combined to produce a continuous color range; hence, yellow combined with blue will give green, whereas combined with red it will give orange. The color printer uses a limited and discontinuous set of components to produce a continuous range of color simply by combining the component in different proportions. I suggest, therefore, that the crucial aspect of semantic encoding of words in an episodic-memory task is analogous to the continuously variable picture that results from the printing and that this should not be confused with the discrete and discontinuous meanings of the individual words presented, on which the episode experienced is based.

Precedents and Consequences

In discussing these views with colleagues who have been more directly involved in current theoretical controversies regarding the processes of retrieval, two questions tend to crop up: first, whether my views are inconsistent with some form of encoding specificity and, second, whether they are testable. I will discuss these questions in turn.

First, the concept of encoding specificity, as with most fruitful concepts, has not remained static over the years. I believe that my views, and the results I cite, are inconsistent with a simple view of encoding specificity that states that cues will enhance recall or recognition only if they are both encoded at the time of learning and reinstated by the experimenter at the time of test. Such a rigid view is not, however, necessitated by more recent versions of encoding specificity (e.g., Flexser & Tulving, 1978; Tulving, 1982). I assume, therefore, that the encoding-specificity theorist does allow some form of mediation via semantic memory between the retrieval cue presented at the test and that presented during learning. There is no reason why such a process cannot be extended to cover the concept of recollection.

Another objection is that I am overgeneralizing existing concepts. Levels of processing was not designed to cover amnesia, and encoding specificity was not intended to refer to semantic memory. I would argue first that virtually all human memory exhibits a blend of semantic and episodic memory. If this distinction is to remain a fruitful one, we shall need to explore both the differences and similarities between semantic and episodic memory more fully than we have so far done. My second justification is that as someone interested in amnesia and word finding, I require theoretical concepts, and I would prefer to have concepts that were also applicable to both episodic and semantic memory and to both patients and normal subjects. The framework I have suggested fulfills this function, and hence, unless it can be shown to be inadequate in other ways, is preferable to more restricted theories.

The second question that may be raised with regard to the proposed framework is that it is not directly testable. Having been brought up on the writings of Popper (1959), I must confess that I have raised this objection, for example, in connection with the concept of levels of processing. I suggest, however, that as a blanket criterion for evaluating all kinds of theory, it is misguided. Theoretical concepts can operate at a whole series of levels ranging from the detailed and precise to the very general. The criteria for evaluating an explanatory concept will depend on the level and purpose for which it is devised. In the present instance the purpose of the theoretical concepts in question is to provide a coherent framework to assist in the understanding of a wide range of phenomena. In this view, a theoretical concept should be evaluated not by whether it approximates some absolute truth but by the extent to which it performs the useful function of interpreting what we already know while facilitating further discovery.

In what was no doubt an attempt to stimulate counter proposals, Tulving and Thomson (1973) claimed, "The formulation of the encoding specificity principle is so general that it covers all known phenomena of episodic memory and retrieval and in a weak sense provides an understanding of them" (p. 370). Unfortunately, the many studies that followed this claim have been concerned not with the adequacy of encoding specificity as a general theory of retrieval but with the specific ingenious but somewhat complex experimental procedure used by Tulving and Thomson in their case against the simple generate–recognize hypothesis. In surveying the 7 years of research on retrieval that have followed Tulving and Thomson's article, one is reminded of a very shrewd comment made by Tulving himself 3 years previously (Tulving & Madigan, 1970): "A phenomenon which should prove to be of great interest to future historians of verbal learning and memory is one that we would like to call the functional autonomy of methods: yesterday's methods have become today's object of study" (p. 442). To be aware of this danger is clearly not sufficient. How can we avoid becoming the prisoners of our methods? I would suggest that one effective antidote is to move out of the laboratory from time to time and attempt to apply our theories and techniques to practical problems. The more bracing air of the outside world may prove too much for some of our laboratory-raised concepts, but I am enough of an optimist to believe that many will survive and flourish.

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